A2 · Unit 5 · WBI15 Respiration, Internal Environment, Coordination and Gene Technology

Cellular Respiration

Cellular respiration is how a cell releases the chemical energy stored in food and captures it as ATP. This page follows the route of glucose through glycolysis, the link reaction, the Krebs cycle and the electron transport chain, then looks at what changes when oxygen runs out.

The big picture

Respiration is not burning. Each step is a small, enzyme-controlled reaction, so energy is released in manageable packets and captured in ATP rather than lost as heat in one go. The respiratory substrate is the molecule that gets broken down, and glucose is the main one. In overall terms, glucose and oxygen give carbon dioxide, water and energy:

$$C6H12O6 + 6O2 \rightarrow 6CO2 + 6H2O + \text{energy (used to make ATP)}$$

Aerobic respiration has two main phases. Glycolysis breaks glucose into a smaller molecule and does not need oxygen. The Krebs cycle needs oxygen, because oxygen is the final acceptor at the end of the electron transport chain. The link reaction connects the two, carrying the product of glycolysis into the mitochondrion.

- **Respiratory substrate:** the molecule that is broken down to release energy, usually glucose.
- **Substrate-level phosphorylation:** ATP formed when a phosphate group is passed directly from an intermediate to ADP.
- **Reduced coenzyme:** NAD or FAD that has accepted hydrogen (and electrons) during a dehydrogenation step.
- **Oxidative phosphorylation:** ATP made using the energy from electrons moving down the electron transport chain.

Stage 1: glycolysis (cytoplasm)

Glycolysis happens in the cytoplasm and does not need oxygen. It starts by adding phosphate groups to glucose, using ATP. This sounds wasteful, but it does two useful things. It makes glucose more reactive so that it can be split, and it stops glucose from leaving the cell because the charged phosphorylated molecule cannot cross the membrane.

The 6-carbon sugar then splits into two 3-carbon triose phosphate molecules (GALP). Each one is oxidised and converted into pyruvate. The oxidation removes hydrogen, which is accepted by NAD to form reduced NAD. Because two ATP were spent at the start and four were made in the later steps, the net gain is 2 ATP per glucose, plus 2 reduced NAD.

KEY IDEA

Per glucose, glycolysis gives a net 2 ATP (4 made, 2 used), 2 reduced NAD and 2 pyruvate. All of this happens without oxygen.

Stage 2: the link reaction (mitochondrial matrix)

Pyruvate is a 3-carbon molecule and must cross the mitochondrial membrane to reach the Krebs cycle. Inside, one carbon is removed as CO2 (decarboxylation), and hydrogen is removed to reduce NAD (dehydrogenation). The 2-carbon remainder is joined to coenzyme A to make acetyl CoA. Two enzymes are needed for these steps, a decarboxylase and a dehydrogenase. Each glucose gives two pyruvate, so the link reaction happens twice per glucose.

Stage 3: the Krebs cycle (mitochondrial matrix)

Acetyl CoA (2C) joins a 4-carbon compound to form a 6-carbon compound, citrate. The cycle then removes carbon as CO2 in two decarboxylation steps, which takes the molecule back down to a 4-carbon compound. That 4-carbon compound is regenerated at the end of the cycle, which is why the cycle can keep turning. Along the way, dehydrogenation steps reduce NAD and FAD, and one step makes ATP directly by substrate-level phosphorylation.

Because each glucose produces two acetyl CoA, the totals per glucose are 4 CO2, 2 ATP, 2 reduced FAD and 6 reduced NAD. Most of the energy from glucose is now held in these reduced coenzymes rather than in ATP.

COMMON MISTAKE

Common mistake: writing that the Krebs cycle makes most of the ATP. It makes only 2 ATP per glucose directly. The bulk of ATP comes later, from the reduced coenzymes passing their electrons to the electron transport chain.

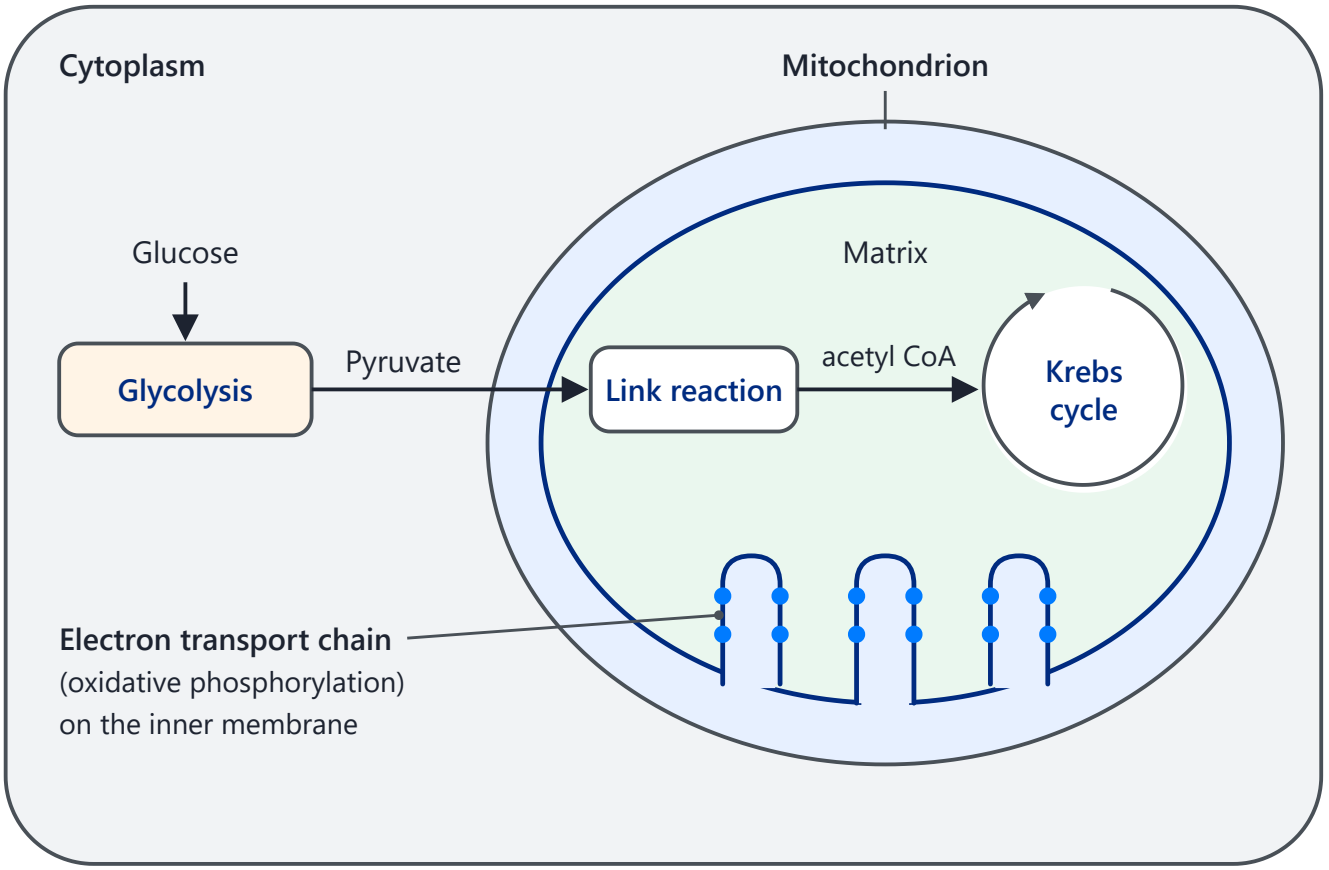
Stage 4: the electron transport chain and oxidative phosphorylation

The electron transport chain (ETC) is a row of electron carriers built into the inner mitochondrial membrane, folded into the cristae. Reduced NAD and reduced FAD release their hydrogen here. The hydrogen splits into electrons and H+ ions. Electrons pass from carrier to carrier, each carrier being reduced when it receives electrons and oxidised when it passes them on. The energy released as electrons move along the chain is used to pump H+ ions from the matrix into the intermembrane space.

This pumping builds up a high concentration of H+ in the intermembrane space, creating an electrochemical gradient. The H+ ions can only diffuse back into the matrix through ATP synthase, and the energy released as they flow through it drives the joining of ADP and inorganic phosphate to make ATP. This coupling of the H+ gradient to ATP synthesis is called chemiosmosis. At the end of the chain, the electrons are taken up by oxygen, which

combines with H+ to form water. Oxygen is therefore the final electron acceptor, and without it the chain backs up and stops.

Reduced NAD becomes oxidised to NAD+ during this process. The regenerated NAD+ returns to glycolysis and the Krebs cycle, so those steps can continue. This regeneration is the reason the whole aerobic pathway depends on oxygen at its far end.



Where each stage of aerobic respiration takes place, from glycolysis in the cytoplasm to oxidative phosphorylation on the cristae.

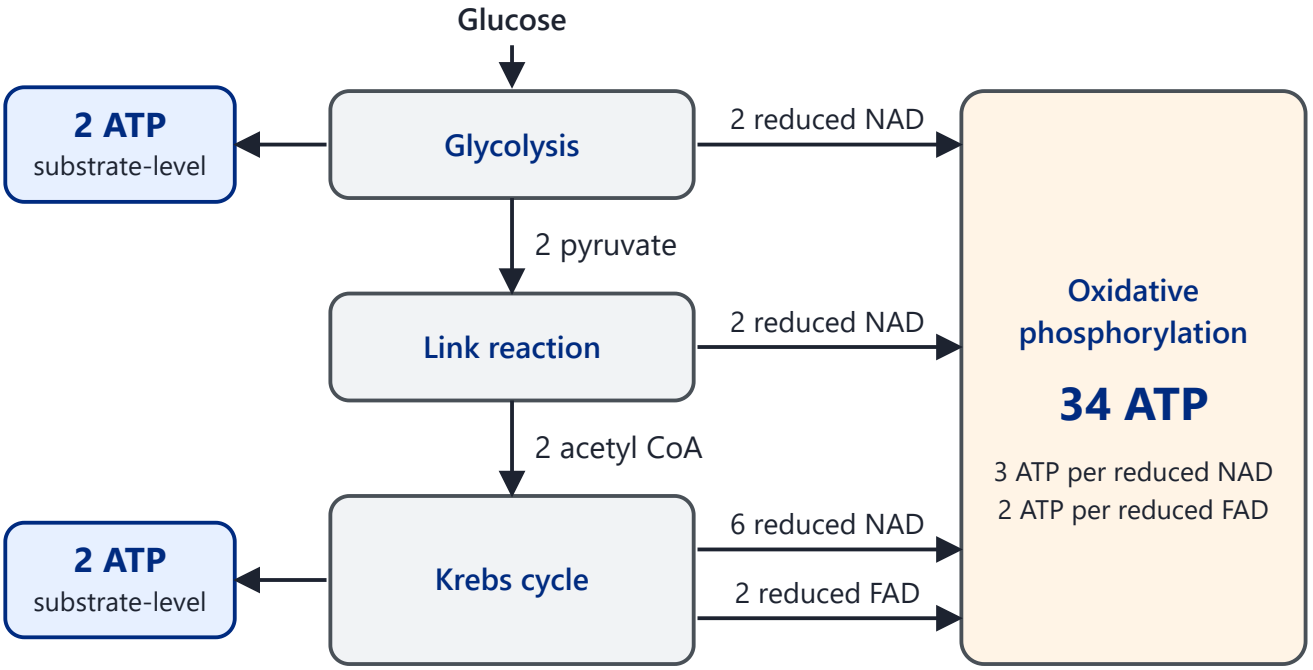
KEY IDEA

Chemiosmosis: H+ are pumped out of the matrix by the energy of electrons, then flow back through ATP synthase, and that flow is what makes ATP.

The model below runs the chain for you, so you can see what changes when oxygen runs out.

Counting the ATP from one glucose

Each reduced NAD is taken to yield about 3 ATP and each reduced FAD about 2 ATP through the ETC. Summing the stages gives the total shown in the standard flow diagram.



Theoretical total: 2 + 2 + 34 = 38 ATP per glucose

ATP and reduced coenzymes produced at each stage, from one glucose molecule. Blue boxes are substrate-level ATP; the orange box is ATP from oxidative phosphorylation.

Theoretical ATP yield per glucose

1. Glycolysis: 2 ATP (substrate level) and 2 reduced NAD. Each reduced NAD gives 3 ATP, so 2 × 3 = 6.
2. Link reaction: 2 reduced NAD (one per pyruvate), so 2 × 3 = 6 ATP.
3. Krebs cycle (two turns): 2 ATP (substrate level), 6 reduced NAD (6 × 3 = 18 ATP) and 2 reduced FAD (2 × 2 = 4 ATP).
4. Total: 2 + 6 + 6 + 2 + 18 + 4 = 38 ATP.

This is the theoretical figure from the rules above. In practice the yield is lower, because some ATP is spent moving molecules across membranes, so check what your mark scheme expects.

Anaerobic respiration and lactate

When oxygen is absent, the electron transport chain cannot pass its electrons on. Reduced NAD cannot unload its hydrogen, so NAD+ is no longer regenerated. The Krebs cycle needs NAD+ and FAD, so it stops. Aerobic respiration therefore stops, but glycolysis does not have to stop: it only needs NAD+ to keep going.

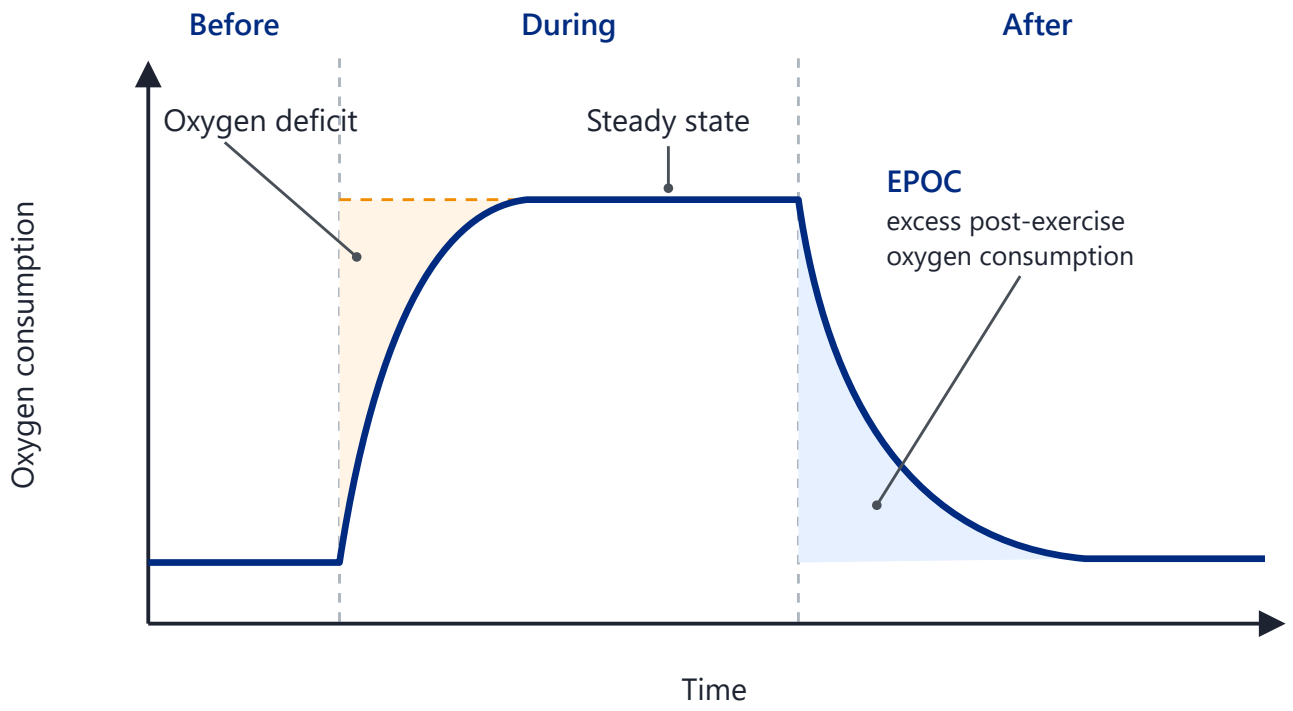
In animal muscle the problem is solved by the pyruvate itself. Pyruvate accepts hydrogen from reduced NAD and is converted into lactate. This regenerates NAD+, so glycolysis can keep producing its net 2 ATP per glucose. No extra ATP comes from the lactate step itself. Yeast and plants do something similar but make ethanol and CO2 instead.

Lactate that builds up must be dealt with later. It is carried in the blood to the liver, where much of it is converted back to pyruvate. That pyruvate is oxidised through the Krebs cycle using extra oxygen, and the rest is turned into glucose and then glycogen for storage in muscle.

Why muscles fatigue

During hard exercise, muscle cells rely heavily on glycolysis and lactate production. The lactic acid lowers the pH of the cell. The H+ ions attach to negatively charged groups in the active sites of enzymes, changing their shape. The substrate can then no longer bind, and the enzyme stops working. This slows the supply of energy and contributes to fatigue and cramp.

The extra oxygen taken in after exercise is called the oxygen debt, or excess post-exercise oxygen consumption (EPOC). It pays for converting lactate back to pyruvate and glucose, and for restoring the body's variables to resting levels.



Oxygen consumption before, during and after exercise. The orange area is the oxygen deficit at the start of exercise; the blue area after exercise is the extra oxygen needed to recover.

EXAM TIP

Exam tip: when a question asks why lactate forms, say that NAD must be regenerated so that glycolysis can continue. Do not say lactate is made to produce energy.

Respiratory substrates and the athlete

The rate of ATP production depends on the substrate and on the size of the inner membrane. Fats store more energy per gram than carbohydrate because they contain a higher proportion of hydrogen. Fat is broken down by lipase into fatty acids and glycerol. Glycerol is phosphorylated and enters glycolysis as GALP. The large amount of hydrogen from fatty acids gives many reduced coenzymes and so a large proton gradient. Protein is used only when carbohydrate and fat are very low; its amino groups are removed (deamination) first.

An athlete's muscle cells often have mitochondria with more cristae. The larger inner membrane carries more electron transport chains and more ATP synthase, so more H+ can be pumped and more ATP made in the same time.

The respiratory quotient

The respiratory quotient (RQ) compares how much CO_2 is released with how much O_2 is used:

$$\text{RQ} = \text{volume of } CO_2 \text{ produced} \div \text{volume of } O_2 \text{ used}$$

An RQ near 1 means carbohydrate is being oxidised, because the O_2 used and CO_2 released match. Fats need much more oxygen for the same CO_2, so their RQ is lower. Protein sits between the two. A value at or above 1 suggests CO_2 is being released without the matching O_2 use, which points to anaerobic respiration.

RQ for oleic acid

1. Equation for aerobic respiration of oleic acid: $C_{18}H_{34}O_2 + 25.5\ O_2 \rightarrow 18\ CO_2 + 17\ H_2O$ + energy.
2. RQ = CO_2 produced ÷ O_2 used = 18 ÷ 25.5.
3. RQ = 0.7, which fits the value expected for a fat.

A fat needs far more oxygen per carbon than glucose, so less CO_2 is released for each O_2 used.

You can measure the RQ of germinating seeds with a simple respirometer, as in the model below.

Quick checks

QUICK CHECK

In which part of the cell does the Krebs cycle take place?

- Cytoplasm
- Mitochondrial matrix
- Intermembrane space
- Cristae of the inner membrane

The Krebs cycle runs in the matrix. The cytoplasm is where glycolysis happens, which is the commonest mix-up. The cristae hold the electron transport chain, not the cycle.

QUICK CHECK

What is the net number of ATP produced by glycolysis per glucose molecule?

- 0
- 2
- 4
- 32

Glycolysis makes 4 ATP by substrate-level phosphorylation but uses 2 at the start to phosphorylate glucose, so the net gain is 2. Quoting 4 is the common error of forgetting the ATP spent.

QUICK CHECK

Why does the electron transport chain stop when oxygen is absent?

- Oxygen is needed to split glucose into pyruvate
- No final electron acceptor, so the electrons and H+ cannot be removed from the chain
- Oxygen is the substrate for ATP synthase
- Lactate blocks the carriers

Oxygen accepts electrons at the end of the chain and combines with H+ to form water. Without it, the carriers stay reduced and reduced NAD cannot be reoxidised, so the chain and chemiosmosis stop.

QUICK CHECK

Why is lactate formed in muscle during anaerobic respiration?

- To store energy as ATP for later use
- To remove carbon dioxide from the cell
- To regenerate NAD+ so that glycolysis can continue
- To pump H+ into the intermembrane space

Pyruvate accepts hydrogen from reduced NAD, which regenerates NAD+ for glycolysis. Lactate itself makes no extra ATP, so the first option is wrong.

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Muscles and the Heart

Movement depends on muscles pulling on bones through tendons, and muscle contraction depends on actin and myosin sliding past each other. The heart is a muscle too, but it contracts on its own rhythm. This page covers the skeleton and muscle structure, the sliding filament mechanism, fibre types, and how the heart's electrical activity produces the cardiac cycle and an ECG.

Bones, cartilage, ligaments and tendons

Each skeletal tissue is built for a particular job. Bone is living tissue. Its osteocytes sit in a hard matrix containing tiny crystals of calcium phosphate and collagen fibres. The calcium phosphate gives compressive strength, while the collagen gives some flexibility so the bone does not snap. Blood vessels running through the bone supply oxygen and nutrients.

Cartilage is a flexible connective tissue made by chondrocytes. It contains no calcium, which is why it is softer than bone. It covers the ends of bones where they meet, reducing friction and absorbing shock.

Ligaments and tendons are both collagen, but they are arranged differently for different jobs. Ligaments join bone to bone. Their fibres are laid in layers, which gives strength with some elasticity, so they hold the joint in place while still allowing movement. Tendons join muscle to bone. Their collagen runs in tight parallel bundles and they are much less stretchy than ligaments. This matters: if a tendon stretched, the muscle's contraction would be used up stretching the tendon instead of moving the bone.

The synovial joint

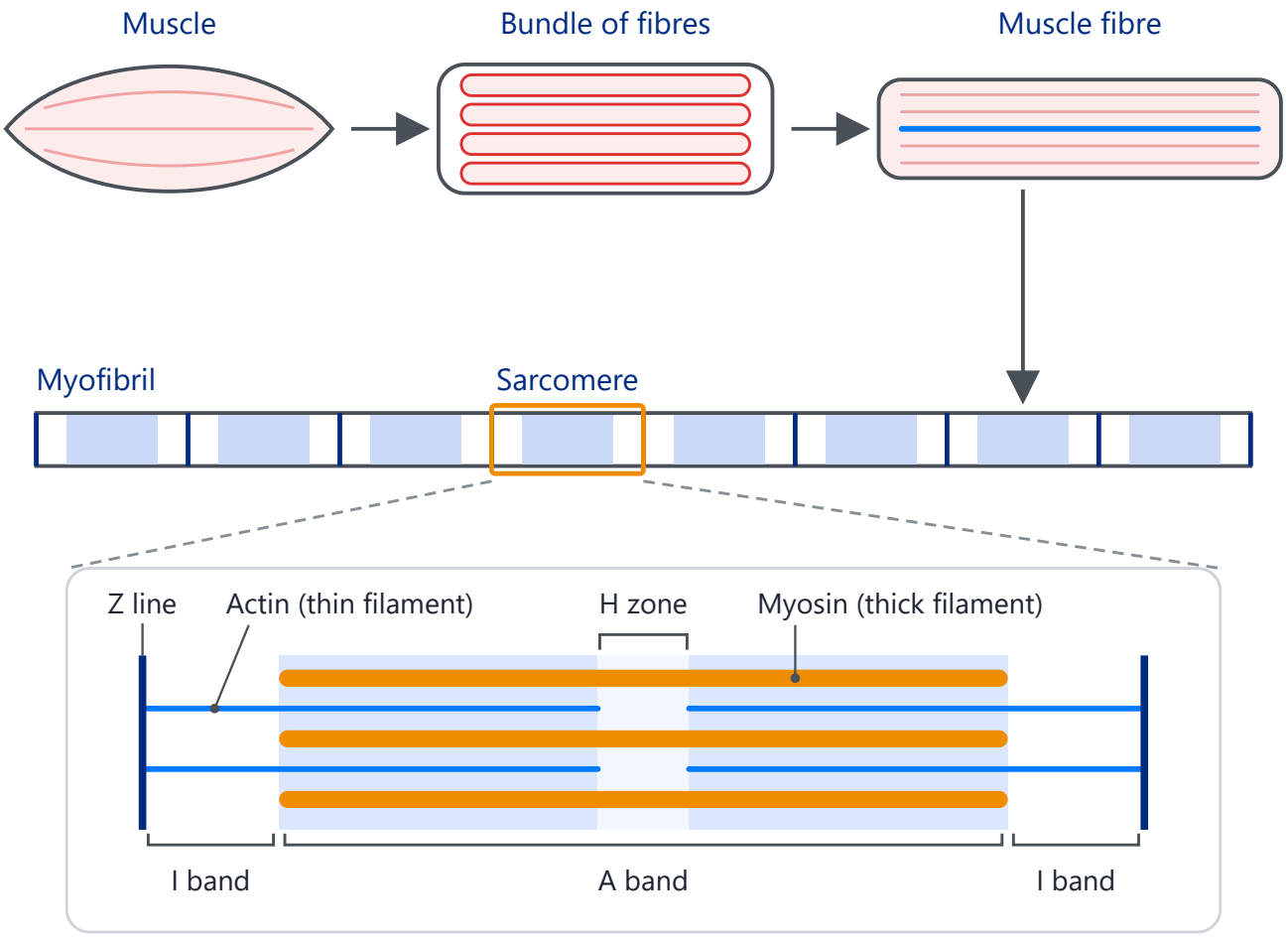
In a synovial joint, the bone ends are capped with smooth cartilage. Ligaments hold the bones together, and a collagen capsule surrounds the joint. The capsule limits how far the joint can move and stops the bones dislocating. Its inner lining, the synovial membrane, secretes synovial fluid into the joint cavity. The fluid is made from blood plasma, and it lubricates the joint, nourishes the cartilage and cuts friction.

Muscle and sarcomere structure

There are three types of muscle: skeletal (striated and voluntary), smooth (unstriated and involuntary, found in gut walls), and cardiac (striated and involuntary). Skeletal muscle is made of very long cells called muscle fibres, bundled together with connective tissue, blood vessels and nerves.

Each fibre contains many myofibrils running parallel to one another. Between the myofibrils is sarcoplasm (the cytoplasm), packed with mitochondria. The sarcoplasmic reticulum is a membrane system that stores calcium ions. The sarcolemma is the cell surface membrane, and it folds inwards at intervals to form T-tubules, which carry the nerve impulse deep into the fibre so the whole fibre contracts together.

A myofibril is made of repeating units called sarcomeres, each one running from one Z line to the next. Two proteins make up the filaments. Actin forms thin filaments, anchored at the Z lines. Myosin forms thick filaments, which are held in the centre of the sarcomere. Because the myosin filaments are thicker, they give the darker A band. The lighter I band contains only actin. The H zone is the central part of the A band where there is myosin but no actin.



From a whole muscle down to the sarcomere, where the thin actin and thick myosin filaments overlap.

Actin, myosin and the regulatory proteins

A thin filament is two strands of actin molecules twisted together, with two other proteins attached. Tropomyosin lies along the actin and blocks the sites where myosin would bind. Troponin sits at intervals along the filament and can bind calcium ions. When calcium binds troponin, troponin changes shape and pulls tropomyosin out of the way, exposing the binding sites.

A myosin molecule is two long polypeptide chains twisted together, with a globular head at one end. The head has two binding sites, one for actin and one for ATP, and it also acts as an ATPase, meaning it splits ATP. Many myosin molecules pack together into a thick filament, with their heads projecting outwards.

The sliding filament mechanism

The sliding filament theory explains contraction without the filaments shortening. Myosin heads pull the actin filaments past them, so the filaments slide over one another and the sarcomere shortens. The cycle needs both calcium and ATP, and it repeats for as long as calcium is present:

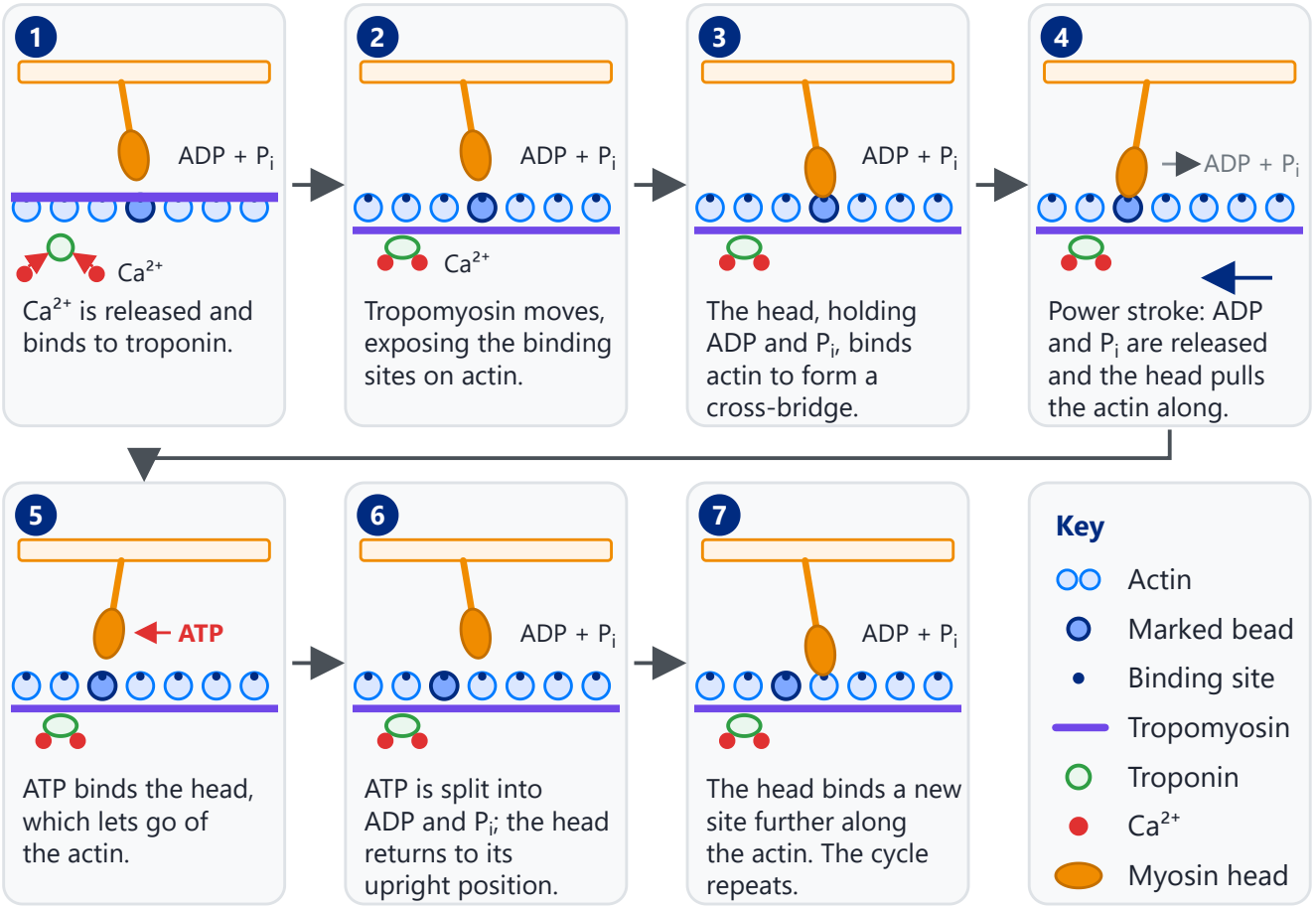
One cycle of cross-bridge movement

1. A nerve impulse reaches the neuromuscular junction, and calcium ions are released from the sarcoplasmic reticulum.
2. Calcium binds troponin, which moves tropomyosin away from the binding sites on actin.

   Without this step, the myosin heads cannot reach actin, so no contraction can start.
3. The myosin head, already carrying ADP and phosphate from the previous cycle, binds to actin and forms a cross-bridge.
4. ADP and phosphate are released, and the head swings forwards (the power stroke), pulling the actin filament along.
5. A fresh ATP molecule binds the head and the bond with actin breaks.
6. The ATPase splits ATP into ADP and phosphate, and the head returns to its upright position, storing energy.

   The energy for the return stroke comes from ATP hydrolysis, so the head can attach again further along the actin.

Calcium has to be pumped back into the sarcoplasmic reticulum when the nerve impulses stop, and that also needs ATP. ATP is therefore needed in three places: to change the shape of the myosin head, to break the cross-bridge, and to remove calcium so the muscle can relax.



The seven stages of cross-bridge cycling, from unblocking the binding site to reattaching further along the actin filament.

COMMON MISTAKE

Common mistake: saying that the actin or myosin filaments themselves get shorter. In contraction the filaments keep their length. The sarcomere shortens because the Z lines move closer together as the actin slides over the myosin, so the H zone and I band narrow.

Use the slider below to contract a sarcomere yourself.

Antagonistic muscles and movement

A muscle can only pull. It cannot push, and it cannot lengthen itself back to its resting length. Movement therefore needs pairs of muscles that work against each other, called antagonistic pairs. A flexor bends a joint and an extensor straightens it. When the biceps (flexor) contracts, the triceps (extensor) relaxes and is stretched. When the triceps contracts, the biceps relaxes. Two muscles in opposition let the joint be controlled in both directions, not just pulled one way.

Fast and slow twitch fibres

Skeletal muscle contains two main fibre types, and the mix in a muscle matches its job. Slow-twitch fibres are aerobic. They are built for steady power over a long period, which suits a marathon runner. Fast-twitch fibres are anaerobic. They give short, explosive contractions, which suits a sprinter or a weightlifter, but they tire quickly.

The differences follow from the way each fibre makes ATP. Slow-twitch fibres have many mitochondria, a large store of myoglobin (an oxygen store that gives them their red colour), and a dense capillary network. Their aerobic respiration keeps lactate low. Fast-twitch fibres have few mitochondria and less myoglobin, but a lot of stored creatine phosphate for quick ATP regeneration and enzymes for anaerobic glycolysis. Because mitochondria take up less space, more myofibrils fit inside, which helps the fibre generate force quickly.

Fast fibres fatigue because anaerobic glycolysis produces lactic acid faster than it can be removed. The pH falls, enzymes stop working well, and contraction fails. This is why a middle-distance or marathon runner relies more on slow fibres and aerobic respiration: lactate would build up too quickly to sustain a sprint pace over a long distance.

Feature	Fast-twitch (white)	Slow-twitch (red)
Respiration	Anaerobic (glycolysis only)	Aerobic (glycolysis and Krebs cycle)
Mitochondria	Few	Many
Myoglobin	Low	High
Capillary density	Low	High
Creatine phosphate	High	Low
Glycogen	High	Low
Fatigue	Fatigues easily	Resists fatigue
Speed of contraction	Fast	Slow

The cardiac cycle

The heartbeat is myogenic. The contraction starts inside the cardiac muscle itself, so the heart does not need an outside nerve to start each beat. The electrical activity that drives it is produced by a specialised group of cardiac cells, which sets the pace and then spreads the signal through the heart in a set order.

The sinoatrial node (SAN), in the right atrium wall, starts each beat and sets the heart rate. Its depolarisation spreads over both atria and makes them contract, pushing blood through the open atrioventricular valves into the ventricles. Non-conducting collagen tissue separates the atria from the ventricles, so the signal cannot pass straight down. It reaches the atrioventricular node (AVN), which delays it briefly. This delay lets the atria finish emptying before the ventricles contract.

From the AVN the impulse travels down the bundle of His and along the Purkinje fibres to the apex of the ventricles. Contraction starts at the bottom and sweeps upwards, squeezing blood towards the arteries. The atrioventricular valves close during this phase, stopping blood from flowing back into the atria. Blood is then forced through the semilunar valves into the aorta (from the left ventricle) and the pulmonary artery (from the right ventricle). In diastole, the ventricles relax, pressure falls, and the semilunar valves close.

The left ventricle is the one that matters for the body's oxygenated blood. Oxygenated blood enters the left atrium from the pulmonary veins, passes into the left ventricle, and is pushed into the aorta by the same wave of contraction.

The graph below follows the pressures on the left side of the heart through one beat.

Cardiac output

Stroke volume is the volume of blood the left ventricle pumps out in one beat. Cardiac output is the volume pumped out per minute, and it depends on both stroke volume and heart rate:

$$\text{cardiac output (cm3 min-1)} = \text{stroke volume (cm3)} \times \text{heart rate (beats min-1)}$$

Change in cardiac output after a swim

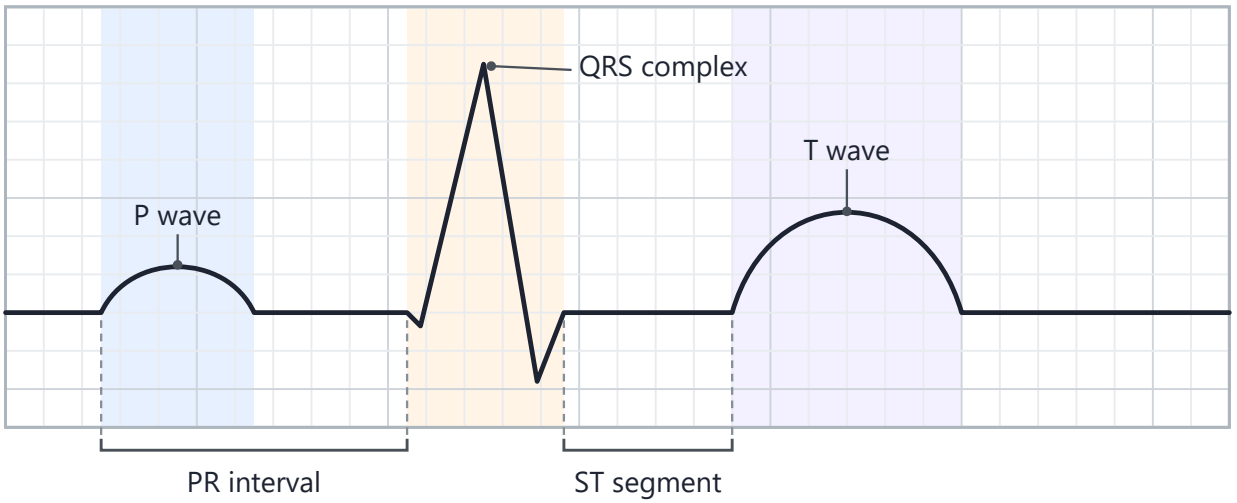
1. At rest: heart rate 65 bpm and stroke volume 70 cm3, so cardiac output = 65 × 70 = 4550 cm3 min-1.
2. After the swim: cardiac output is 10 800 cm3 min-1 and heart rate is 120 bpm.
3. Change in cardiac output = 10 800 − 4550 = 6250 cm3 min-1.
4. Stroke volume after the swim = 10 800 ÷ 120 = 90 cm3.

Rearranging the same equation gives the missing variable. Both rising heart rate and rising stroke volume account for the increase.

Reading an ECG

Each contraction generates small electrical currents in the body fluids around the heart. Electrodes placed on the skin pick these up, and the recording is an electrocardiogram (ECG). An ECG has characteristic waves:

- **P wave:** depolarisation of the atria, so atrial systole.
- **PR interval:** the time for the impulse to pass from the SAN, across the atria, through the AVN.
- **QRS complex:** ventricular depolarisation, so ventricular systole.
- **ST segment:** the period when the ventricles are contracted and then begin to relax.
- **T wave:** repolarisation of the ventricles, so ventricular diastole.

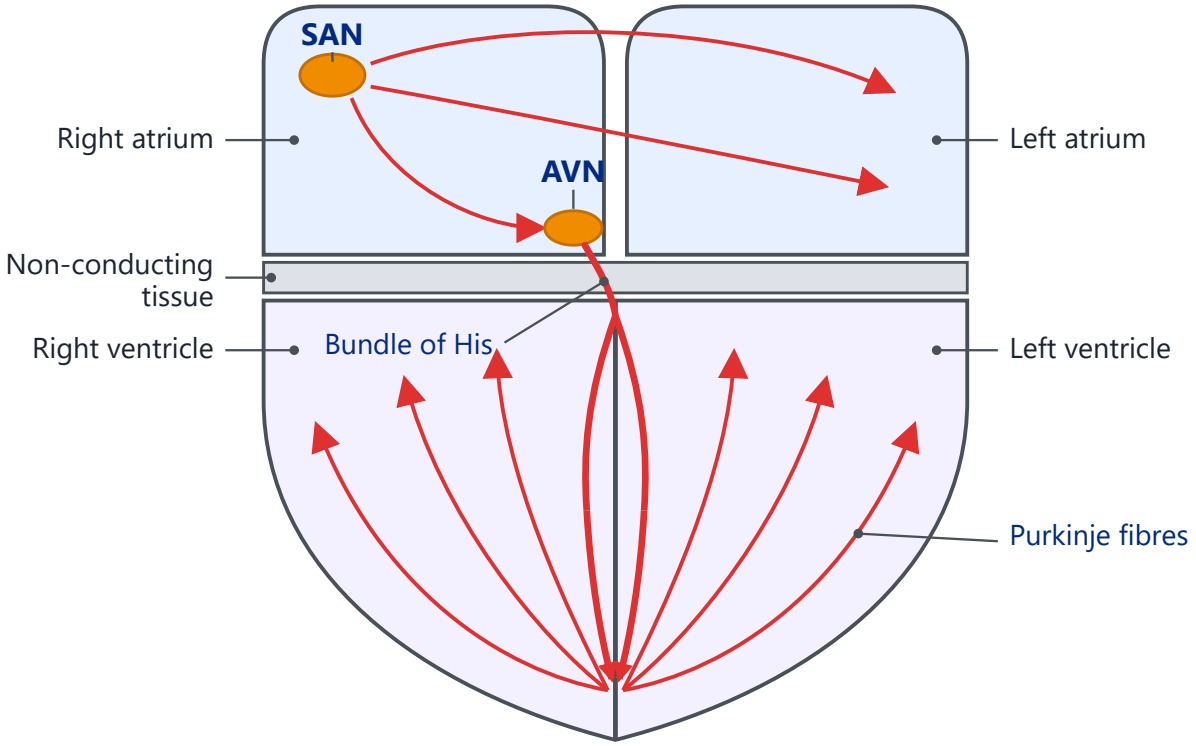


P wave
Atria depolarise:
atrial systole

QRS complex
Ventricles depolarise:
ventricular systole

T wave
Ventricles repolarise:
ventricular diastole

One normal heartbeat on an ECG, with the waves matched to atrial and ventricular activity.



The conduction pathway through the heart: the SAN starts the signal, which spreads over the atria to the AVN, then passes down the bundle of His and out along the Purkinje fibres. The red arrows show its direction.

The heart rate comes from the time between successive beats. One beat is measured from one P wave to the next, or from one QRS complex to the next, and the rate is the number of beats divided by the time taken.

Heart rate from an ECG

1. If the P-to-P interval is 0.8 s, one beat takes 0.8 s.
2. Beats per minute = 60 ÷ 0.8 = 75 bpm.

Dividing 60 by the time for one beat converts it to beats per minute.

ECGs help diagnose heart problems by showing abnormal rhythms. A very irregular trace (fibrillation) can cause chest pain, fainting or death. Tachycardia means a fast resting rate, around 120 bpm, so the heart may not be pumping efficiently. An ectopic beat is an extra beat started early, which shows up as an unusual wave out of the normal sequence. A clinician compares the trace with a normal one, and an abnormal trace may point to cardiovascular disease or to a faulty AVN.

EXAM TIP

Exam tip: when describing the SAN to AVN delay, say that it lets the atria empty before ventricular contraction starts. Give the purpose, not just the fact of a delay.

Quick checks

QUICK CHECK

Which structure sets the rate of the heartbeat?

- Atrioventricular node
- Sinoatrial node
- Bundle of His
- Purkinje fibres

The SAN starts each beat and sets its rate. The AVN only delays the signal, and the bundle of His and Purkinje fibres carry it to the ventricles. Choosing the AVN is the usual mistake, because it is the node that gets the most attention in the pathway.

QUICK CHECK

Why must the myosin head detach from actin during the cycle?

- To release calcium back into the sarcoplasmic reticulum
- So that tropomyosin can block the binding site again
- So that a fresh ATP can bind and the head can reattach further along actin
- To shorten the A band

Detachment needs a fresh ATP. Its hydrolysis then returns the head to its upright position, allowing a new attachment further along actin. Calcium is pumped back by ATP, and the A band does not change length during contraction.

QUICK CHECK

A heart beats at 60 beats per minute with a stroke volume of 80 cm3. What is its cardiac output?

- 4800 cm3 min-1
- 140 cm3 min-1
- 480 cm3 min-1
- 1.3 cm3 min-1

Cardiac output = stroke volume × heart rate = 80 × 60 = 4800 cm3 min-1. Adding the two values (140) confuses the formula with a sum, and 480 comes from a slipped digit. Dividing by 60 gives a per-second figure, not per minute.

QUICK CHECK

On an ECG, which wave corresponds to ventricular depolarisation?

- P wave
- QRS complex
- T wave
- PR interval

The QRS complex is ventricular depolarisation, which leads to ventricular systole. The P wave is atrial, the T wave is ventricular repolarisation (relaxation), and the PR interval is a time, not a wave.

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Control of the Internal Environment

Body cells work only if the conditions around them stay within narrow limits. This page explains homeostasis and feedback, then applies them to three examples: how the heart rate and cardiac output change, how breathing responds to carbon dioxide, and how body temperature is held near 37 °C.

Homeostasis and dynamic equilibrium

Homeostasis is the maintenance of a stable internal environment. Its key feature is that conditions are not fixed but kept within a range around a set point. The body's temperature, blood glucose, blood pH and blood gas levels all keep moving slightly, and the control systems correct them as they drift. This is called dynamic equilibrium, and it is why the word "balance" is misleading: the system is always correcting.

Every control system has the same parts. A receptor detects a change, a control centre compares the signal with the set point, and an effector carries out a response that moves the variable back. The receptor and effector are often in different places, so the signal has to travel by nerve or by hormone.

Negative and positive feedback

Negative feedback is the usual form of control. A change in one direction triggers a response in the opposite direction, so the variable is pulled back towards the set point. This keeps conditions within a narrow range, and it is why most homeostatic systems oscillate around a value rather than drifting away from it. Thermoregulation and blood pressure control both work this way.

Positive feedback does the reverse. The response reinforces the original change, pushing the variable further from normal until something stops it. Blood clotting is the standard example. Activated platelets release a chemical that activates more platelets, and the process escalates until the wound is sealed. Positive feedback is useful when a fast, decisive response is needed, but it cannot be left running indefinitely.

> **KEY IDEA**
>
> Negative feedback keeps a variable near its set point. Positive feedback amplifies a change until a stop signal ends it.

Controlling heart rate and cardiac output

Heart rate is set by the sinoatrial node, but the rate the SAN fires at is adjusted by the cardiovascular control centre (CCC) in the medulla oblongata. The CCC works unconsciously and sends signals along two sets of nerves. The parasympathetic nerve (part of the vagus) releases acetylcholine onto the SAN and AVN, slowing the rate. The

sympathetic nerve, the cardiac accelerator nerve, releases noradrenaline and speeds the rate and the force of contraction.

The CCC receives its information from receptors. Baroreceptors in the carotid sinus and aortic arch detect blood pressure, and chemoreceptors in the carotid and aortic bodies detect oxygen, carbon dioxide and pH. The effects are summarised below:

Stimulus detected	Receptor	Nerve used by CCC	Effect on SAN and heart rate
High blood pressure	Baroreceptors	Parasympathetic (acetylcholine)	Fewer impulses; heart rate falls
Low blood pressure	Baroreceptors	Sympathetic (noradrenaline)	More impulses; heart rate rises
High O2, low CO2, high pH	Chemoreceptors	Parasympathetic	Heart rate falls
Low O2, high CO2, low pH	Chemoreceptors	Sympathetic	Heart rate rises

Adrenaline adds a hormonal route to the same result. Stress or exercise stimulates the hypothalamus, which signals the adrenal medulla through the sympathetic system. The adrenal medulla releases adrenaline into the blood. Adrenaline binds receptors on the SAN and increases its firing, so heart rate rises before the body is even short of oxygen. This is the fight or flight response. At the same time it raises blood glucose by stimulating the liver, releases fatty acids from fat stores, and redirects blood away from the gut and skin towards the muscles and heart.

The same reflexes respond to changes in venous return. When more blood returns to the right atrium, its walls stretch and send more impulses to the cardioacceleratory centre (the Bainbridge reflex). This increases heart rate and helps keep venous pressure from building up.

Stroke volume

Stroke volume depends on three things: the end diastolic volume (EDV), the average aortic pressure, and the strength of ventricular contraction. The most important is EDV, which is the blood left in the ventricles at the end of diastole. The more blood returns to the heart, the more the cardiac muscle fibres are stretched. Stretched fibres contract more strongly, so more blood is ejected per beat. This relationship is called the Frank-Starling mechanism. Because venous return controls EDV, a fast return of blood (as in sprinting) raises stroke volume as well as heart rate.

How exercise raises cardiac output

1. Muscles contract harder, so more blood returns to the heart through the veins.
2. The increased return raises EDV, stretching the ventricle walls.

   A more stretched fibre contracts with greater force, so stroke volume rises.
3. Adrenaline and the sympathetic nerve raise heart rate at the same time.

4 Cardiac output (stroke volume × heart rate) therefore rises from both sides.

The model below shows both changes together.

Ventilation and breathing rate

Breathing is controlled by two centres in the medulla: an inspiratory centre and an expiratory centre. Stretch receptors in the walls of the bronchial tree provide negative feedback. When the lungs inflate, the receptors fire and send impulses to the expiratory centre through the vagus nerve. This stops inspiration. The internal intercostal muscles then contract, the external ones relax, and the elastic recoil of the lungs pushes air out. Once the bronchi are no longer stretched, the receptors stop firing, the expiratory centre switches off, and a new inspiration starts.

Carbon dioxide is the main signal for changing the rate. Exercise produces CO_2, which dissolves in the blood to form carbonic acid and lowers pH. The chemoreceptors in the carotid and aortic bodies detect this and send more impulses to the inspiratory centre. The diaphragm and external intercostals then contract more often and more deeply. Breathing out removes CO_2 by diffusion down its concentration gradient, which is why holding the breath lets CO_2 build up in the blood and drives the urge to breathe.

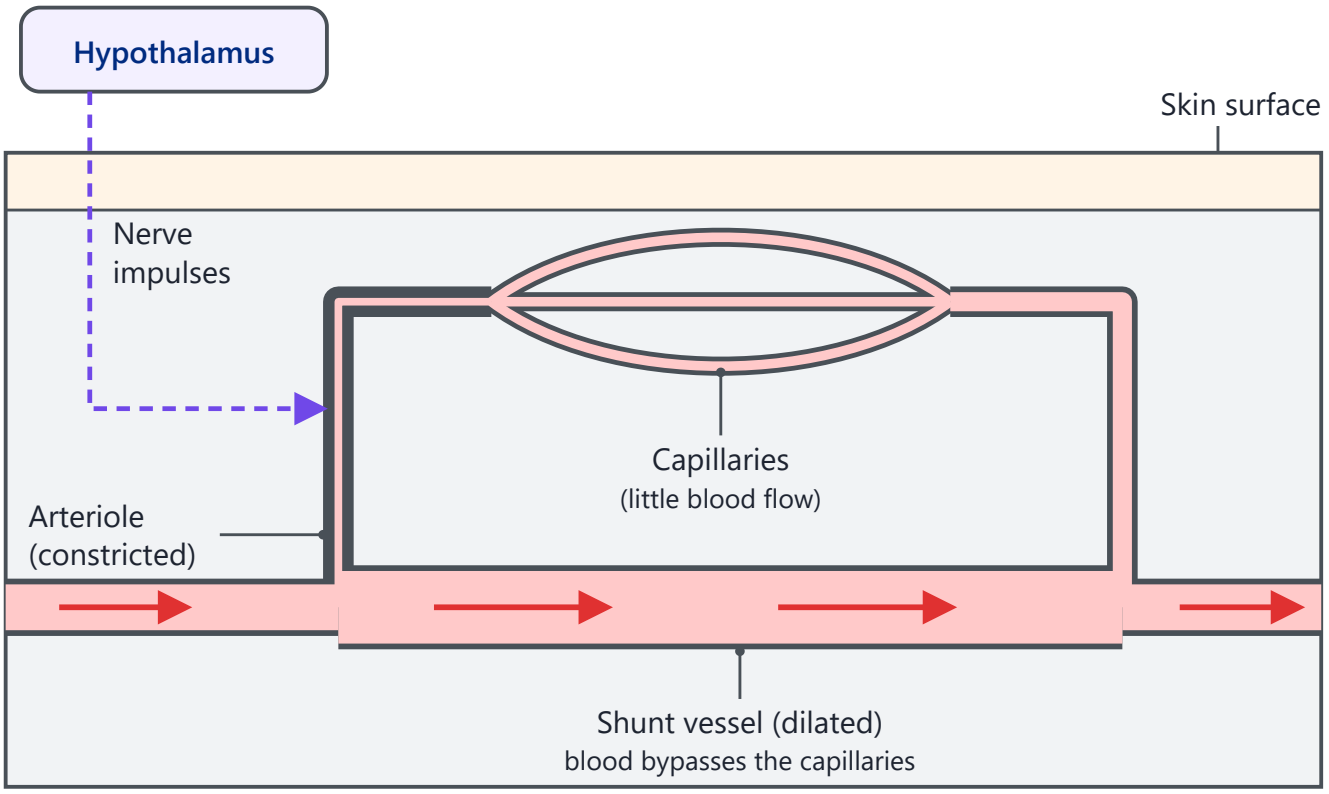
Exercise raises the demand for oxygen, so breathing rate and depth and heart rate all increase. Faster blood flow through the alveolar capillaries keeps the diffusion gradient steep, so more oxygen enters the blood per second.

Thermoregulation

Core temperature is held near 37 °C, the set point. Thermoreceptors in the hypothalamus are very sensitive to small changes in the temperature of the blood passing through it. The hypothalamus contains two centres. The heat loss centre is activated when temperature rises, and the heat gain centre when it falls. Both work through negative feedback, so the body oscillates slightly around the set point.

When the body is too warm, the skin arterioles dilate and shunt vessels constrict, so more blood flows close to the surface and more heat is lost by radiation, conduction and convection. Sweat glands are stimulated, and evaporation of the sweat removes heat. Hairs lie flat, so less air is trapped. Metabolic rate and muscular activity fall.

When the body is too cold, the skin receptors and the hypothalamus trigger the opposite changes. Arterioles constrict and shunt vessels dilate, so less blood reaches the skin and less heat is lost. Sweating is inhibited. Erector pili muscles raise the hairs to trap a layer of insulating air. Shivering produces heat through muscle contraction, and the hormones adrenaline and thyroxine increase the metabolic rate.



In cold conditions the arteriole constricts and the blood is diverted through the shunt vessel, which keeps heat in the core.

Response	Lowering temperature	Raising temperature
Sweating	Increased; evaporation cools	Inhibited; less heat loss
Hairs	Lie flat (erector muscles relax)	Stand up (erector muscles contract, trap air)
Skin arterioles	Dilate; more heat by radiation	Constrict; less heat lost
Muscle activity and metabolism	Reduced	Shivering; adrenaline and thyroxine raise metabolism

COMMON MISTAKE

Common mistake: saying that the body "cools itself by sweating" in cold conditions. Sweating only happens when the set point is exceeded, and it is suppressed when the body is cold.

The model below shows this negative feedback at work.

Quick checks

QUICK CHECK

What is the main purpose of the Bainbridge reflex?

- To lower heart rate after exercise
- To raise heart rate when venous return increases, limiting build-up of venous pressure
- To constrict the arterioles in the skin
- To stop breathing during exercise

Increased venous return stretches the atrial walls, which triggers faster heart rate through the sympathetic system. The first option describes the baroreceptor response instead, which is the commonest confusion between the two reflexes.

QUICK CHECK

Which change would NOT help a person who has become too cold?

- Hairs stand up to trap air
- Arterioles in the skin constrict
- Shivering increases
- Sweat glands are stimulated to cool the skin

In cold conditions sweating is inhibited, because evaporation would lose heat. The other three responses all conserve or produce heat.

QUICK CHECK

What is the effect of a high blood CO_2 level on breathing?

- Breathing rate falls to save oxygen
- Chemoreceptors signal the inspiratory centre, so breathing rate and depth increase
- Stretch receptors stop the expiratory centre
- The diaphragm relaxes completely

Raised CO_2 lowers blood pH, which the chemoreceptors detect. They send more impulses to the inspiratory centre, so breathing speeds up and deepens. Stretch receptors work on lung inflation, not on CO_2 levels.

QUICK CHECK

Which example is a positive feedback mechanism?

- Control of body temperature by sweating
- Baroreceptor control of blood pressure
- Platelets releasing a chemical that activates more platelets during clotting
- Breathing rate controlled by stretch receptors

In clotting, each change reinforces the next, which is what positive feedback does. The other three options all bring a variable back towards a set point, which makes them negative feedback systems.

A2 · Unit 5 · WBI15 Respiration, Internal Environment, Coordination and Gene Technology

Nervous System and Coordination

Coordination means one part of the body responds to a change detected in another part. The nervous system does this with electrical impulses that travel along neurones, and the hormonal system does it with chemical messengers in the blood. This page covers neurone structure, the nerve impulse and synapses, the reflex arc, the effects of drugs on synapses, and how hormones act.

Organisation of the nervous system

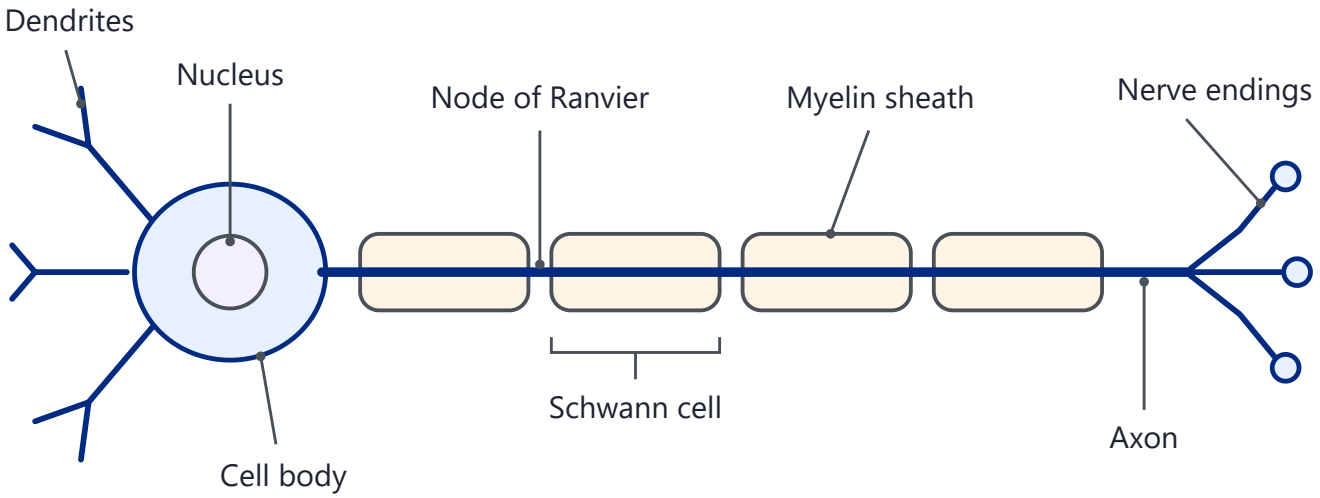
The nervous system is a network of interconnected neurones. The central nervous system (CNS) is the brain and spinal cord, where information is processed. The peripheral nervous system (PNS) connects the CNS to the rest of the body. Sensory neurones carry impulses from receptors into the CNS, and motor neurones carry impulses from the CNS to effectors such as muscles and glands.

The motor part of the PNS splits into two divisions. The somatic division controls voluntary skeletal muscle. The autonomic division controls involuntary activity in smooth muscle, cardiac muscle and glands. Within the autonomic system, the sympathetic division prepares the body for action by raising metabolic rate (fight or flight). The parasympathetic division does the opposite and conserves energy (rest and digest).

Neurone structure

A neurone has a cell body containing the nucleus. The cell body is compact and enclosed by a membrane, and it contains Nissl granules, which are rough endoplasmic reticulum and ribosomes used to make neurotransmitters. Short branching dendrites bring impulses towards the cell body. A single long axon carries impulses away from it.

Many axons are wrapped in a myelin sheath, made by Schwann cells that coil their membrane around the axon many times. The myelin is a lipid-rich insulator. The gaps between adjacent Schwann cells are called nodes of Ranvier. Myelin matters because the current cannot leak through it, so depolarisation only happens at the nodes. The impulse therefore jumps from node to node. This is called saltatory conduction, and it is much faster than conduction along an unmyelinated axon. Breakdown of the myelin sheath causes multiple sclerosis, which slows or blocks transmission.

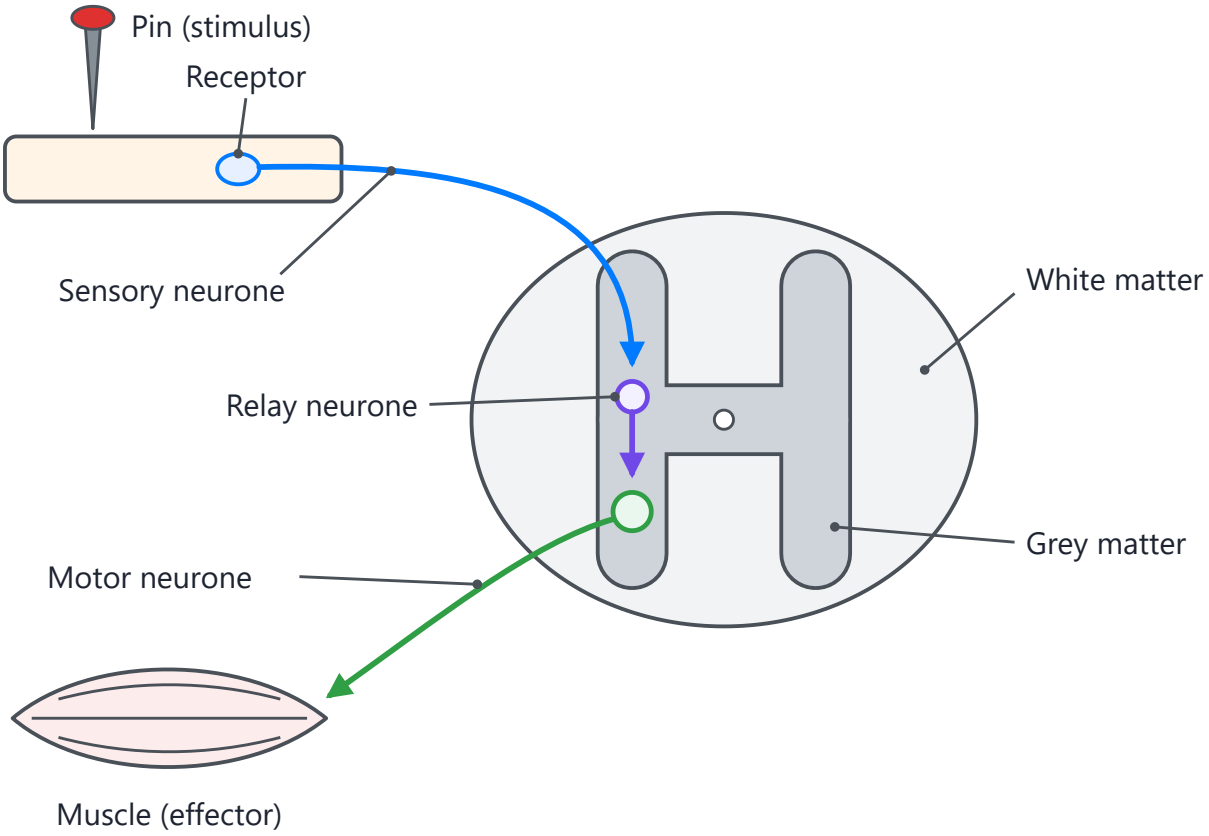


A motor neurone with its dendrites, cell body, myelinated axon (Schwann cells and nodes of Ranvier) and nerve endings labelled.

Types of neurone and the reflex arc

The three neurone types differ in where they sit and how they are built. Sensory neurones have a cell body off to one side of the fibre, with a long dendron running to the receptor and a short axon into the CNS. Motor neurones have a cell body at one end, many short dendrites, and a long myelinated axon ending at the effector. Relay neurones lie entirely within the CNS, have short axons and are not myelinated. They pass impulses from sensory to motor neurones.

A reflex arc is the simplest nervous response. It is the pathway from a receptor to an effector without involving conscious parts of the brain. If the pathway goes through the spinal cord but not the brain, it is called a spinal reflex. The stages are: a receptor detects the stimulus; a sensory neurone carries the impulse into the spinal cord; a relay neurone passes it across a synapse; a motor neurone carries it to the effector; the effector responds. Reflexes are fast because they bypass the brain. In the spinal cord, the cell bodies of relay neurones lie in the grey matter, while the myelinated axons that form the white matter carry impulses up and down the cord.



A spinal reflex arc, from a skin receptor to a muscle, with the white and grey matter of the cord labelled.

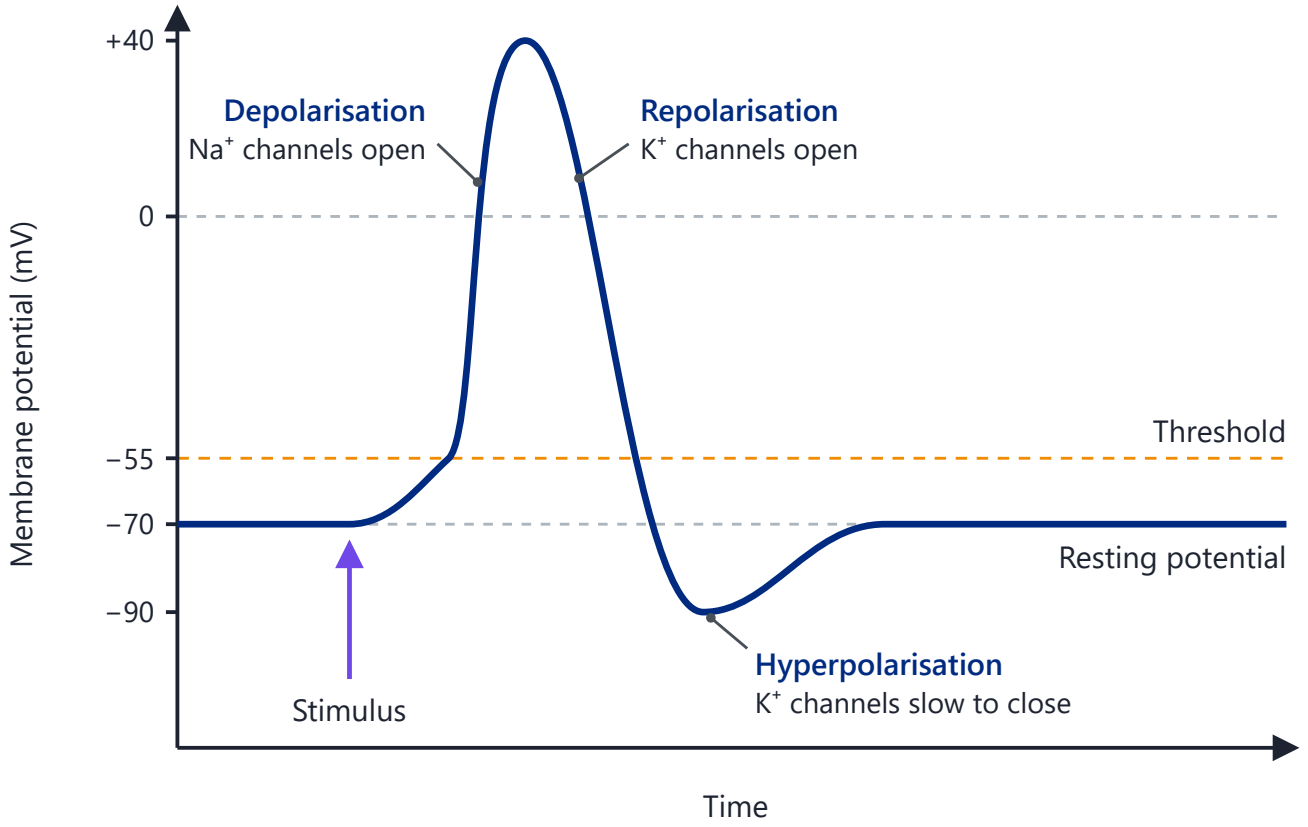
The resting potential

A neurone at rest is polarised: the inside is about −70 mV relative to the outside. This is produced by the sodium-potassium pump, which uses ATP to move three Na+ out of the cell for every two K+ moved in. The pump builds concentration gradients for both ions. The membrane is much more permeable to K+ than to Na+ at rest, so K+ leaks out down its gradient. This leaves the inside negative, which pulls K+ back in. At −70 mV the two forces balance, so there is no net movement of potassium. Na+ cannot leak back in quickly because few sodium channels are open.

The pump is essential because without it the gradients would run down and the membrane potential would disappear.

The action potential

When a stimulus arrives, the membrane becomes more permeable to Na+, and voltage-gated sodium channels open. Na+ rushes in, and the inside becomes positive. This is depolarisation. If the change reaches the threshold (about −55 mV), more sodium channels open in a positive feedback loop, and the potential rises rapidly to about +30 to +40 mV. The sodium channels then close and inactivate, while voltage-gated potassium channels open more slowly. K+ leaves the cell, which repolarises the membrane. The K+ channels close slowly, so too many K+ leave and the membrane briefly becomes hyperpolarised, reaching about −90 mV. The pump and leak channels then restore the resting potential.



The change in membrane potential during an action potential, with the phases of Na+ and K+ channel activity labelled.

The action potential, step by step

1. Resting potential: the inside of the axon is about −70 mV compared with the outside. The sodium-potassium pump moves 3 Na+ out for every 2 K+ in, and K+ leaks out through open potassium channels. The voltage-gated Na+ and K+ channels are closed.

2. Depolarisation: a stimulus opens some sodium channels and Na+ diffuses in, so the inside becomes less negative. If the potential reaches the threshold of about −55 mV, voltage-gated sodium channels open.

   If threshold is not reached, no action potential is produced and the membrane returns to rest.

3. Na+ floods in, which opens more voltage-gated sodium channels (positive feedback). The potential rises rapidly to about +30 to +40 mV, so the inside is now positive.

4. Repolarisation: the sodium channels close and are inactivated, and voltage-gated potassium channels open. K+ diffuses out, so the inside becomes negative again.

5. Hyperpolarisation: the potassium channels are slow to close, so too many K+ leave and the potential falls below the resting value, to about −90 mV.

6. Refractory period: while the sodium channels are inactivated, no new action potential can start (absolute refractory period). As they recover, only a larger than normal stimulus can start one (relative refractory period). The potassium channels close, and the pump and leak channels restore the resting potential of −70 mV.

The refractory period keeps impulses separate and makes them travel in one direction only.

- **Threshold:** the membrane potential that must be reached before an action potential is generated, about −55 mV here.
- **All-or-none:** once threshold is reached, every action potential has the same size, whatever the strength of the stimulus.
- **Absolute refractory period:** the sodium channels are inactivated, so no new action potential can start.
- **Relative refractory period:** a larger than normal stimulus can trigger another action potential, because the membrane is still hyperpolarised.

The refractory period has a useful consequence. An action potential cannot travel backwards, because the part of the membrane just behind the wave is still refractory. The impulse therefore moves in one direction only.

A stronger stimulus does not produce a larger action potential. It produces more action potentials per second. Stronger stimuli depolarise the receptor more, so they reach threshold more often, and the impulse frequency rises. Two inputs that are too small to reach threshold alone can still fire a neurone if they add together. Spatial summation combines inputs from several neurones at the same time. Temporal summation combines rapid repeated inputs from one neurone.

You can test threshold, all-or-none and the refractory period on the model below.

Synapses and acetylcholine

Neurones are separated from the next cell by a synapse, a small gap called the synaptic cleft. When an action potential reaches the presynaptic knob, voltage-gated calcium channels open and Ca2+ flows in. The calcium triggers synaptic vesicles to fuse with the presynaptic membrane and release their neurotransmitter into the cleft. The transmitter diffuses across and binds to receptors on the postsynaptic membrane. In a cholinergic synapse, the transmitter is acetylcholine, and its receptors open sodium channels, so the postsynaptic cell depolarises.

The synapse then has to be cleared quickly, or the signal would keep firing. The enzyme acetylcholinesterase breaks acetylcholine into acetate and choline. These products are taken back into the presynaptic knob and rebuilt into acetylcholine, using ATP from the many mitochondria in the knob. The synapse is one-way: only the presynaptic knob holds vesicles, and only the postsynaptic membrane has receptors. A single neurone can also receive inputs from many others (convergence), and one neurone can send signals to many others (divergence).

Neurotransmitters can be excitatory, making the postsynaptic membrane more likely to fire, or inhibitory, making it less likely to fire.

The model below shows how the effects of several synapses add together.

Drugs that act at synapses

Drugs change nerve transmission by copying, blocking or altering the steps at a synapse. An agonist increases the effect of a transmitter, for example by stopping its breakdown or by blocking its reuptake. An antagonist reduces the effect, for example by blocking the receptor or by stopping release.

- **Nicotine:** mimics acetylcholine and binds its receptors, opening the channels. With repeated exposure the receptor becomes desensitised and stops responding for a time, which is why it does not simply keep stimulating the synapse.
- **Lidocaine:** a local anaesthetic that blocks voltage-gated sodium channels in sensory neurones. Without sodium entry, no action potential is set up, so no pain signal reaches the brain.
- **Cobra alpha toxin:** binds acetylcholine receptors at the neuromuscular junction and blocks them, so muscles are not stimulated. Paralysis of breathing muscles can cause death.
- **L-DOPA:** used to treat Parkinson's disease, which involves the death of dopamine-producing neurones. Dopamine itself cannot cross the blood-brain barrier, but L-DOPA can. It is converted into dopamine in the brain, which restores transmission and reduces tremor and stiffness.
- **MDMA (ecstasy):** blocks the reuptake of serotonin into the presynaptic neurone and causes it to be released. Serotonin stays in the synapse for longer, so the postsynaptic neurones keep being stimulated.

EXAM TIP

Exam tip: for any drug question, name the step it affects (release, binding, breakdown or reuptake) and say whether the net effect is more or less transmission. Vague answers like "it affects the brain" will not score.

Hormones and coordination

Hormones are chemical messengers released into the blood by endocrine glands, which have no ducts. Each hormone acts only on target cells that carry its receptor. Hormones are needed in small amounts and act more slowly than nerve impulses, but their effects can last much longer. Exocrine glands, by contrast, release their secretions through ducts. The pancreas is both: its islets of Langerhans are endocrine, and its acinar cells are exocrine.

Most hormonal control uses negative feedback. A rising level of the stimulus triggers hormone release, and the hormone then lowers the stimulus, which reduces the release. The pituitary gland is controlled by the hypothalamus. Neurosecretory cells in the hypothalamus release releasing factors into a special blood supply that reaches the anterior pituitary, and they store and release hormones such as ADH and oxytocin from the posterior pituitary.

The way a hormone acts depends on whether it can cross the cell membrane. Peptide hormones are too large and polar to enter cells, so they bind surface receptors and trigger a signalling cascade inside. Steroid hormones are

lipid-soluble, so they pass through the membrane and bind receptors inside the cell. The steroid–receptor complex then acts as a transcription factor, switching specific genes on or off so that the cell makes different proteins.

Transcription factors are central to this. They are proteins that bind to the promoter region of a gene and help RNA polymerase assemble at the start of transcription. Activators assist the assembly and switch the gene on. Repressors block it and keep the gene off. Some transcription factors are present in all cells, while others are made only in particular cell types or at particular stages of development.

Quick checks

QUICK CHECK

What is the main reason the impulse travels in one direction along an axon?

- Dendrites block the impulse from going back
- The myelin sheath only covers one side of the axon
- The membrane behind the impulse is refractory, so it cannot depolarise again
- Calcium ions move the impulse forwards

The region behind the wave is still repolarising or hyperpolarised, and its sodium channels are inactivated. It cannot be depolarised again, so the wave can only move forwards. Calcium is involved in synaptic transmission, not in conduction along the axon.

QUICK CHECK

Which change would make a postsynaptic neurone MORE likely to fire?

- Blocking acetylcholine receptors
- Stopping calcium entry into the presynaptic knob
- Preventing acetylcholinesterase from breaking down acetylcholine
- Increasing the rate at which the pump moves K+ into the cell

Acetylcholinesterase clears acetylcholine from the cleft. If it is blocked, acetylcholine stays longer and keeps stimulating the postsynaptic membrane, so firing is more likely. Blocking receptors or calcium entry would reduce transmission.

QUICK CHECK

Why does a stronger stimulus not change the size of the action potential?

- Sodium channels are always fully open
- The action potential is all-or-none, so once threshold is reached its size is fixed; stronger stimuli only increase frequency
- The myelin sheath limits the size
- Potassium ions cancel out the extra sodium

An action potential either happens or it does not. A stronger stimulus reaches threshold more often, so more impulses are fired per second, but each one is the same size.

QUICK CHECK

Why can L-DOPA treat Parkinson's disease when dopamine cannot?

- L-DOPA stops the death of dopamine neurones
- Dopamine is destroyed in the stomach but L-DOPA is not
- L-DOPA crosses the blood-brain barrier and is converted into dopamine inside the brain
- L-DOPA blocks the reuptake of dopamine

Dopamine cannot cross the blood-brain barrier, so giving it directly does not help. L-DOPA crosses, and it is turned into dopamine in the brain, restoring transmission. It does not stop the loss of neurones.

A2 · Unit 5 · WBI15 Respiration, Internal Environment, Coordination and Gene Technology

Gene technology

Gene technology cuts out a gene, joins it to a carrier molecule and places it in a new cell, so the cell makes a protein it could not make before. This topic covers the enzymes that make recombinant DNA, how genes are copied and separated, the vectors used to carry them, the production of GMOs in plants, animals and microorganisms, gene therapy, microarrays and bioinformatics, and the risks and benefits of GMOs.

1. Recombinant DNA and GMOs

Recombinant DNA is DNA made by joining pieces from two or more different sources. An organism that contains recombinant DNA is called a genetically modified organism (GMO), or a transgenic organism when the new gene comes from a different species. The genetic code is shared by almost all living things, so a gene from a human can be read and expressed by a bacterium, a plant or a goat, provided it is delivered with the right control sequences.

The process follows a standard sequence. First the gene of interest is identified and isolated. Then it is cut out, copied, joined to a vector, delivered into a host cell, and the cells that have taken it up are identified and cloned. Each step depends on enzymes that either cut DNA or join it together, so those enzymes are the foundation of the technique.

2. Restriction enzymes and ligase

Restriction endonucleases (restriction enzymes) cut the sugar–phosphate backbone of DNA at specific base sequences. Each enzyme recognises a particular short sequence and cuts at that site. Many of these recognition sequences read the same in both directions, so the enzyme cuts both strands in matching places. Some enzymes cut straight across both strands, but many cut in a staggered way and leave short single-stranded overhangs called sticky ends.

Sticky ends are useful because they are complementary. A sticky end cut by a given enzyme will pair with any other sticky end cut by the same enzyme, through hydrogen bonds between complementary bases. This means a gene cut with an enzyme can be inserted into a plasmid cut with the same enzyme. The pairing is only temporary, though, because hydrogen bonds are weak. To make the join permanent, DNA ligase is needed.

DNA ligase forms phosphodiester bonds between the sugar and phosphate groups of adjacent nucleotides, sealing the sugar–phosphate backbone on both strands. Together with the complementary base pairing, it produces a continuous, double-stranded molecule.

Making a recombinant plasmid

1. Cut the gene out of the source DNA with a restriction enzyme, leaving sticky ends.

   The enzyme gives a defined cut, so the gene has known ends.

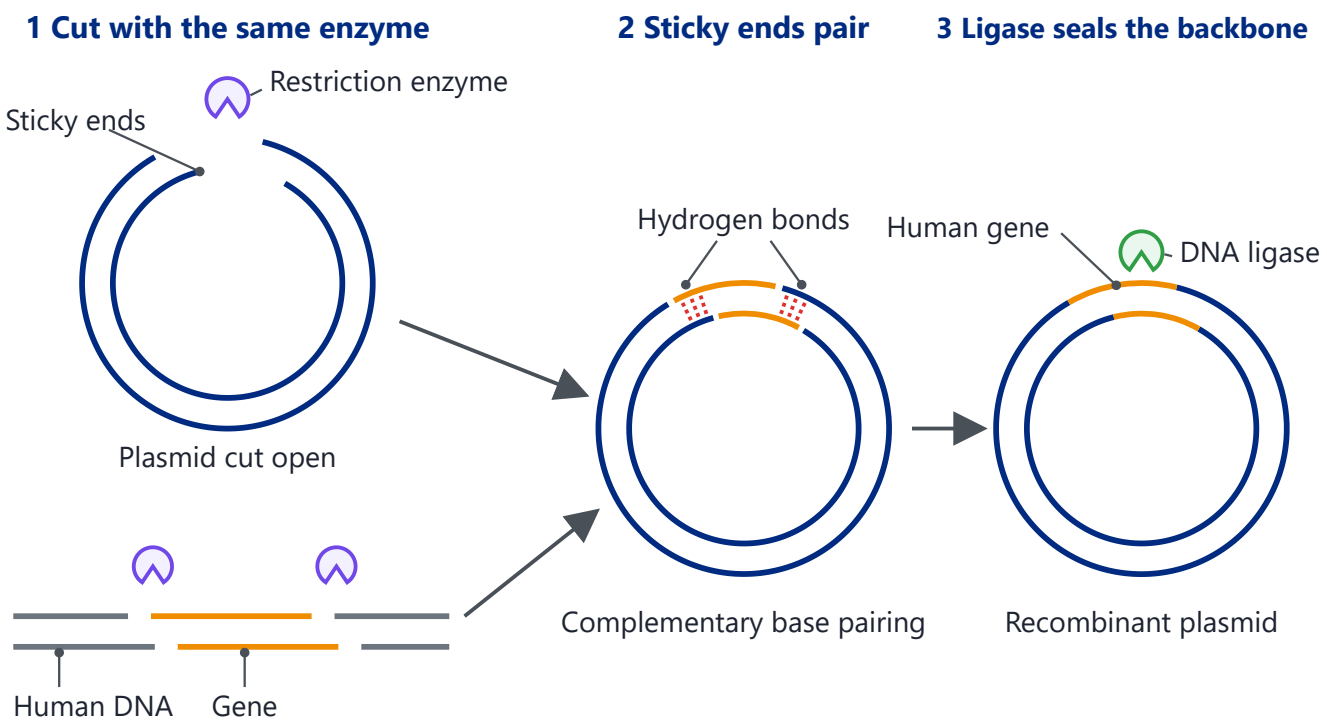
2. Cut the plasmid open with the same restriction enzyme.

   The plasmid's sticky ends then match the gene's sticky ends.

3. Mix the gene and the plasmid so that complementary sticky ends pair through hydrogen bonds.

4. Add DNA ligase to form phosphodiester bonds and seal the backbone.

   Without ligase, the join would fall apart because hydrogen bonds are weak.



Forming a recombinant plasmid. The same restriction enzyme cuts both the gene and the plasmid, so their sticky ends match before ligase seals them.

3. Making copies of genes

A gene can be obtained in three ways. It can be cut from a chromosome with restriction enzymes, it can be copied from mRNA, or it can be made from nucleotides. When the gene is taken from mRNA, the enzyme reverse transcriptase builds a single-stranded DNA copy called cDNA. This copy has no introns, because introns are removed from the mRNA before it leaves the nucleus, so it is the version the host cell can express most easily. DNA polymerase then makes the cDNA double-stranded.

The polymerase chain reaction (PCR) makes many copies of a short DNA sequence in vitro. Each cycle has three steps. The DNA is heated to about 95 °C so the two strands separate. The temperature is lowered so that short primers can bind to the ends of the target sequence. The temperature is then raised to around 72 °C, where a

heat-stable DNA polymerase builds a new complementary strand from each primer. Because each cycle doubles the number of copies, the amount of DNA grows exponentially, and a few cycles can produce millions of copies.

4. Gel electrophoresis

Gel electrophoresis separates DNA fragments by size. The DNA is loaded into wells in an agarose gel and an electric current is applied. DNA has a negatively charged phosphate backbone, so the fragments move towards the positive electrode. The gel acts like a sieve: small fragments pass through the gel pores quickly, and large fragments are slowed down. After a set time the fragments have spread into bands, with the smallest furthest from the wells. Comparing the bands with a size marker allows the sizes of the fragments to be estimated, and the bands can be stained to make them visible.

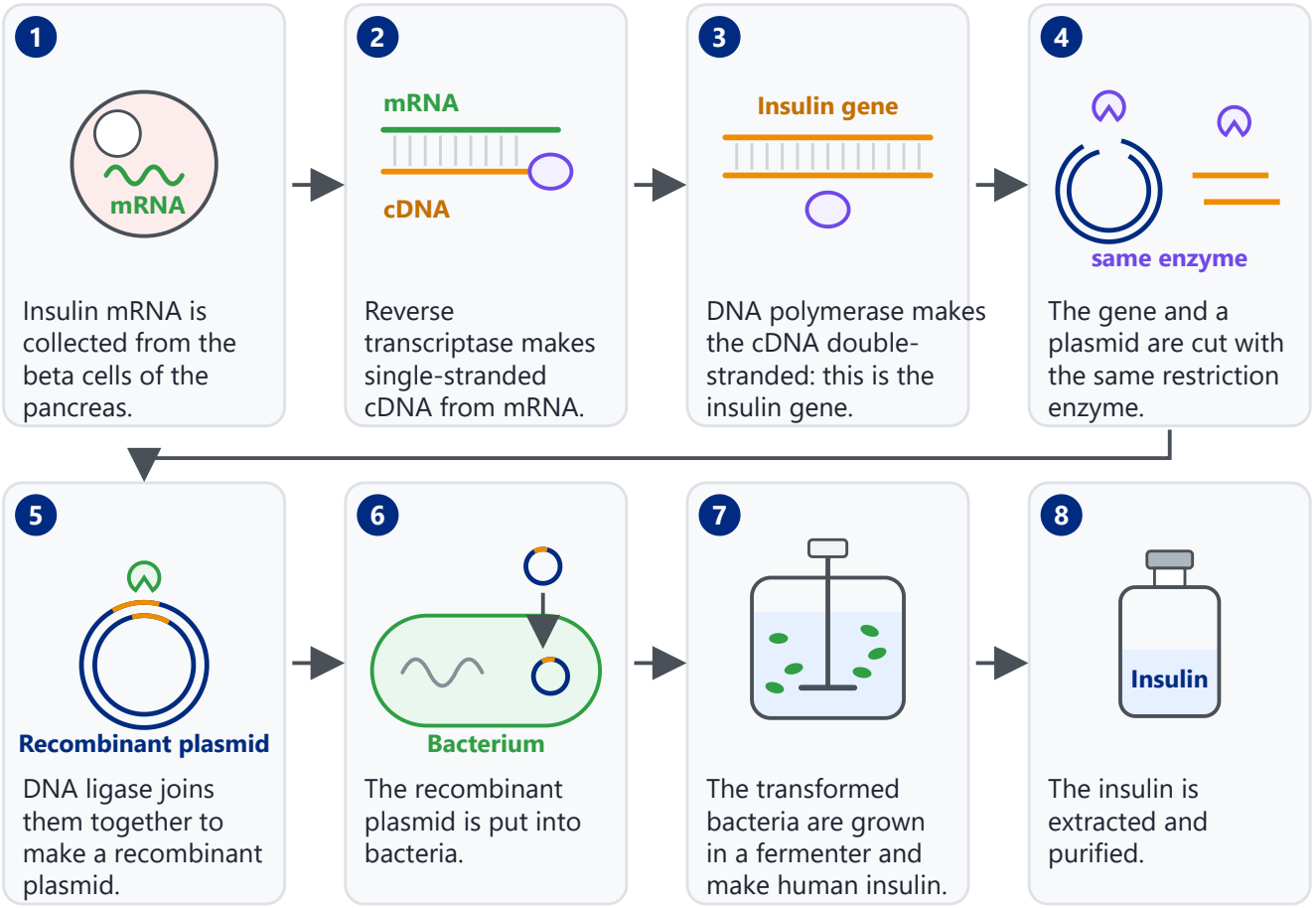
5. Vectors and ways of delivering genes

A vector carries the gene into the host cell. Bacterial plasmids are the usual vector for bacteria. They are small circular pieces of DNA that replicate separately from the chromosome, they are easy to take up and to purify, and they often carry a marker gene, such as an antibiotic resistance gene or a gene for a fluorescent protein. Scientists use the marker to identify cells that have taken up the plasmid. Plasmids are cut open with the same enzyme used to cut the gene, so the recombinant plasmid contains the desired gene, the marker and the bacterial replication origin.

Several methods deliver genes into cells. A gene gun fires DNA coated onto tiny gold or tungsten pellets into cells at high speed. A harmless virus can be used if the gene for viral replication has been removed and replaced with the desired gene; the virus then infects the cell and delivers the gene. Liposomes are small fat vesicles that wrap around the DNA, fuse with the cell membrane and release the DNA into the cytoplasm. They cause fewer side effects than viruses. Microinjection uses a fine glass micropipette to inject DNA directly into the nucleus of a cell. It is precise, but very few cells can be treated, so it is slow and inefficient.

6. Microorganisms and insulin

Microorganisms are cheap to grow and reproduce quickly, so a transferred gene is copied very rapidly when the microorganism is allowed to replicate under ideal conditions. Human insulin is a classic example. The gene is isolated from the beta cells of the pancreas, using the mRNA: reverse transcriptase makes cDNA, and DNA polymerase makes it double-stranded. The cDNA is cut with restriction enzymes, joined to a cut plasmid with ligase, and the recombinant plasmid is inserted into bacteria. The transformed bacteria are grown in a large fermenter, where they produce the human protein, which is then extracted and purified.



Producing human insulin with bacteria. The bacteria are the factory, so the gene is copied and expressed in every cell that grows.

KEY IDEA

Bacteria are used for insulin because they have no introns to remove, they replicate fast and they can be grown in large fermenters. A gene from a eukaryote that has been copied from mRNA is therefore easy for a bacterium to express.

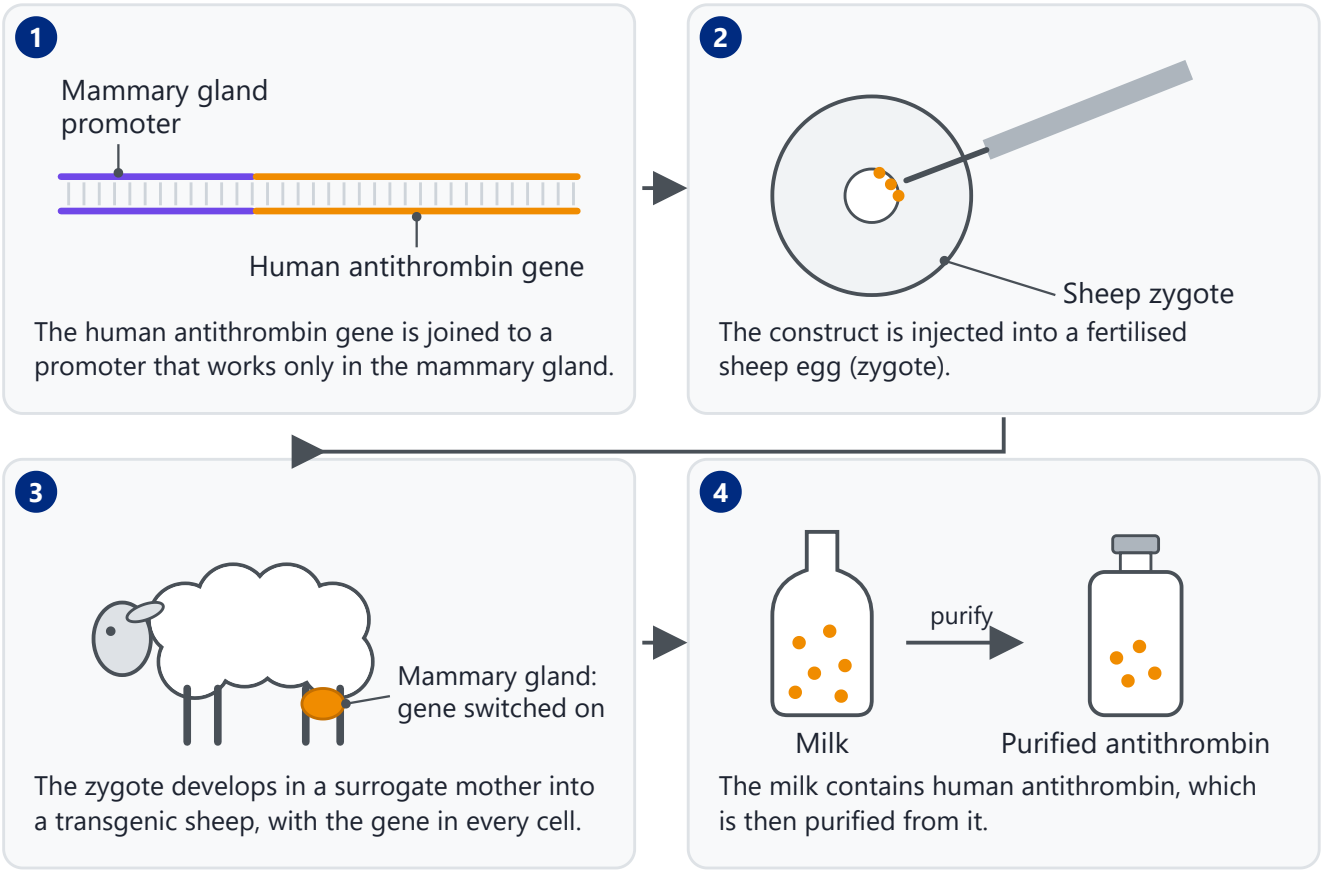
7. Transgenic plants

Transgenic plants are used mainly to produce crops with new traits and, in some research, edible drugs such as vaccines. The most common vector is the Ti plasmid from the soil bacterium Agrobacterium tumefaciens. In nature, this bacterium infects wounds in plants and transfers part of its plasmid, the T-DNA, into the plant chromosome, causing a tumour. Scientists remove the tumour-causing genes, insert the desired gene into the T-DNA, and let the bacterium carry it into plant cells. Infected cells are then cultured, and a whole plant is regenerated from a single cell, so every cell of the new plant carries the foreign gene.

Edible vaccines work in a similar way. A gene from a human pathogen is inserted into a vector and introduced into plant cells, which grow into leaf segments that produce the antigen. Eating the plant material triggers an immune response, so no refrigeration or injection is needed.

8. Transgenic animals

A transgenic animal has a genome that has been changed to carry genes from another species. Transgenic animals are used as disease models and as producers of medicines. To make a transgenic animal that makes a human protein in its milk, scientists isolate the human gene, add a promoter sequence that switches the gene on only in the mammary gland of a lactating female, and inject the construct into a fertilised egg. The egg is placed in a surrogate mother. The resulting animal carries the gene in every cell, and the protein is purified from its milk. The number of animals can be increased by breeding or cloning.



A transgenic animal producing a human protein in its milk. Only the mammary gland switches on the human gene.

COMMON MISTAKE

The promoter is essential. Without it, the gene might be expressed in every tissue, and the protein would not appear in the milk. When you describe the production of a protein in milk, name the mammary-specific promoter.

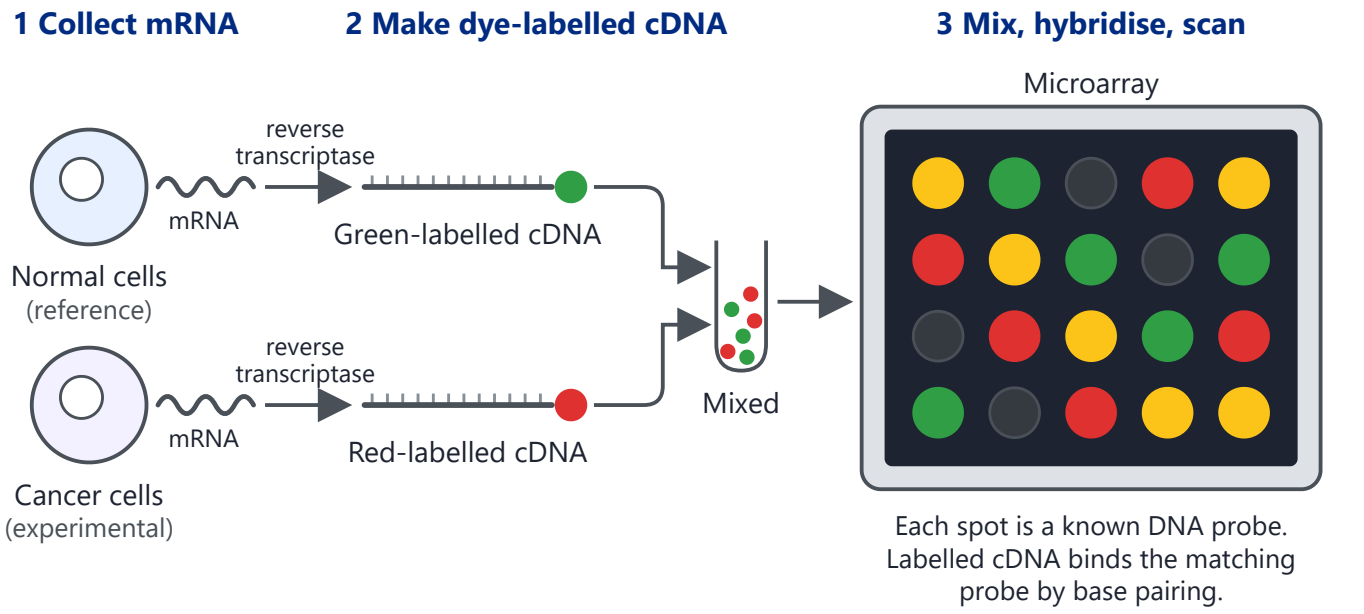
9. Gene therapy

Gene therapy replaces a defective allele with a normal one, so that the genotype and phenotype of the target cells are corrected. There are two forms. Somatic cell therapy treats cells that are not involved in reproduction, for

example airway epithelial cells in cystic fibrosis. The normal gene is introduced by a vector such as a virus or a liposome, and it is transcribed and translated into normal protein. Because the treated cells are replaced over time, the treatment has to be repeated. Germ-line therapy repairs the gene in a fertilised egg, so the repaired gene is copied into every daughter cell by mitosis and is passed on to the next generation. It is inherited, which is why it is more controversial.

10. Microarrays and bioinformatics

A DNA microarray is a glass or plastic slide carrying thousands of spots, each containing a known single-stranded DNA sequence. It shows which genes are active in a cell, because active genes make mRNA. To use a microarray, mRNA is collected from two samples: a reference sample of known gene sequence, such as normal cells, and an experimental sample, such as cancer cells. Reverse transcriptase converts each mRNA into cDNA, and each sample is labelled with a different fluorescent dye, green for the reference and red for the experimental sample. The labelled samples are mixed and applied to the slide, where the cDNA binds to the matching probes by complementary base pairing. The slide is then scanned, and the colour of each spot shows the relative expression. Yellow means both samples express the gene equally, red means the experimental sample expresses it more, and green means the reference expresses it more.



The workflow for a microarray experiment. The colour of each spot is a summary of which genes are switched on in each sample.

Bioinformatics is the development of software, computing tools, algorithms and statistical models to store, organise and analyse raw biological data. Microarrays produce enormous amounts of data, so bioinformatics is needed to find patterns in it. For example, comparing gene expression in healthy and Parkinson's disease tissue would involve many genes at once, and bioinformatics identifies which differences are significant and which are noise.

The same tools can be used to work out gene profiles, to find causes of disease and to assess whether a treatment is working.

11. Risks and benefits of GMOs

GMOs offer clear benefits. Crops can be modified to give higher yields or more nutrition, which can help reduce famine and malnutrition. Pest resistance means fewer pesticides are needed, which lowers costs and reduces environmental damage. Industrial enzymes can be made cheaply in GMOs. Disorders can be treated with human proteins, which are safer and more effective than animal proteins. Vaccines made in plant tissue do not need refrigeration, which helps their distribution in areas without it, and drugs made in transgenic plants or animals may be cheaper to produce.

The risks are also real. Genetic pollution can occur when GM plants cross with conventional plants of the same species or with wild relatives, and the natural genome can be disturbed. Herbicide-resistant crops could interbreed with wild plants and create superweeds, which would force farmers to use more selective herbicides and damage food chains. The use of antibiotic resistance genes as markers may contribute to resistance in microbes. Some GM plants may not produce fertile seed, so farmers must buy new seed from biotechnology companies each year. Some people worry about the long-term and unforeseen effects of GMOs, the possibility that a modified plant or animal could be toxic or cause allergies, and the ethical view that modifying animals purely for human benefit is wrong.

When asked to discuss GM crops, give points for and against and link each to its evidence. For example, fewer pesticides are needed and fewer pests survive on resistant crops, but pollen transfer may spread the gene to wild relatives. A balanced answer states the benefit, then the risk, then which one a given situation makes more important.

12. Quick checks

QUICK CHECK

What is the role of DNA ligase in making recombinant DNA?

- It cuts the DNA at specific sequences
- It copies mRNA into cDNA
- It forms phosphodiester bonds to seal the sugar–phosphate backbone
- It separates the two strands of DNA

DNA ligase joins the backbone of the gene to the plasmid, making phosphodiester bonds. Restriction enzymes do the cutting, reverse transcriptase makes cDNA and heat separates strands in PCR. Hydrogen bonds between sticky ends hold the pieces together only temporarily.

QUICK CHECK

In a microarray experiment, what does a red spot show?

- The gene is expressed equally in both samples
- **The gene is expressed more in the experimental sample**
- The gene is expressed more in the reference sample
- The gene is not expressed in either sample

The experimental sample is labelled red, so a red spot means more of that mRNA was present in it. Yellow means equal expression, and green means the reference sample expressed the gene more. A spot with no colour would mean no expression.

QUICK CHECK

Why is a mammary-specific promoter included in a construct used to make a transgenic animal?

- To stop the gene being copied during replication
- To remove introns from the gene
- **So that the human gene is switched on only in the mammary gland**
- To make the fertilised egg divide faster

The promoter controls where the gene is expressed. Placing the human gene behind a mammary-specific promoter means the protein is made in the milk, where it can be collected, and not in every tissue.

QUICK CHECK

In gel electrophoresis, why do DNA fragments move towards the positive electrode?

- DNA is positively charged
- The gel has a positive charge
- The phosphate backbone of DNA is negatively charged
- Small fragments are attracted to the bands

The phosphate groups in the DNA backbone carry a negative charge, so the fragments are pulled towards the positive electrode. The gel then separates them by size, with the smallest fragments travelling furthest.